Architecture-dependent effects of imbalance correction in melanoma classification: fairness, calibration, and explanation analysis

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Abstract

Deep learning models for melanoma detection are trained on imbalanced, predominantly light-skinned data. We ask whether modern imbalance correction changes the ranking of CNN and transformer architectures, whether calibration explains the change, and how the resulting models behave across skin tone and lesion contrast. We benchmark ResNet-50, EfficientNet-B0, and ViT-S/16 on HAM10000 under three pre-declared conditions: cross-entropy with standard augmentation, focal loss with class-balanced sampling, and cross-entropy with strong augmentation. Each condition uses three seeds, a lesion-aware split, and a validation-selected threshold at 95% specificity. Focal loss with class-balanced sampling raises ViT-S/16 sensitivity by 8.2 points and lowers ResNet-50 by 11.7 and EfficientNet-B0 by 7.9, which reverses the point-estimate ranking. The paired within-architecture intervals include zero, but the difference-in-differences between each CNN and the ViT excludes zero. The AUC ranking is stable, and focal loss roughly triples expected calibration error, which identifies calibration as a mechanism behind the thresholded reversal. A lightness-stratified analysis shows that the dark stratum has the highest false positive rate at L^* cutoffs of 55, 58, and 60, while the dark AUC deficit is present only at the strictest cutoff. Adjusted models show opposite associations: darker skin is associated with more false positives and lighter skin with more false negatives, and lesion-skin contrast moves both error types in opposite directions. Across all three architectures, Grad-CAM attribution mass inside the lesion is lower on darker images, and false negatives have lower mass than false positives, which explains the weak pooled mass-error correlation. External validation on 1,280 HIBA dermoscopic images reproduces the interaction: focal loss raises ViT-S/16 sensitivity by 9.4 points and lowers the two CNNs by 7.6 and 14.3, while the CE-baseline sensitivity remains 16 to 18 points below HAM10000. HIBA contains no Fitzpatrick IV melanomas and no Fitzpatrick V-VI images. The results show that imbalance correction has architecture-dependent effects, that calibration tracks the sensitivity reversal, and that fairness in this setting is error-type-specific rather than a single sensitivity gap.

Keywords: melanoma classification, class imbalance, fairness, calibration, explainable artificial intelligence, skin tone

1 Introduction

Melanoma is the most lethal form of skin cancer, and survival depends on early detection. Deep learning models have reached dermatologist-level accuracy on public dermoscopic benchmarks such

as HAM10000 and the ISIC challenges [35, 6]. Three problems remain.

Public datasets are imbalanced. Benign lesions outnumber melanomas by roughly ten to one, and a classifier that predicts the majority class can look accurate while missing cancers. Focal loss and class-balanced sampling are common remedies [21], but their effect is often reported for one architecture and one dataset.

The datasets are also skewed toward lighter skin. Fitzpatrick skin types V and VI are rare in public dermoscopic data, and models trained on lighter skin can lose sensitivity on darker skin [15, 10, 30]. The data needed to measure the gap are themselves scarce.

Deep models are opaque as well. Gradient-based attribution methods such as Grad-CAM and Integrated Gradients can show which pixels drive a prediction [32, 33], but most dermatology papers use them as illustrations. They are rarely turned into quantitative measurements that can be tested across skin tone groups.

Most studies examine one of these problems at a time. Architecture comparisons often apply a single imbalance correction without checking whether it is neutral across architectures, and fairness studies report group-level performance without connecting it to attribution behaviour. This study connects the three in one benchmark.

We study HAM10000 [35] with a binary melanoma endpoint and three architectures: ResNet-50, EfficientNet-B0, and ViT-S/16. We use a lesion-aware split, three seeds per condition, and a validation-selected threshold at 95% specificity. The test set is touched once per run.

Our contributions are:

1. A controlled benchmark of three architectures under three pre-declared conditions: CE baseline, focal loss with class-balanced sampling, and CE with strong augmentation.

2. Evidence that focal loss with class-balanced sampling improves ViT-S/16 but reduces the sensitivity of both CNNs, which reverses the point-estimate ranking at 95% specificity. The paired within-architecture intervals include zero, while the difference-in-differences between each CNN and the ViT excludes zero. The AUC ranking is stable.

3. A calibration analysis that shows focal loss with class-balanced sampling roughly triples expected calibration error, which identifies calibration as a mechanism behind the sensitivity reversal.

4. A lightness-stratified fairness analysis at three cutoffs. The dark stratum has a higher false positive rate at every cutoff, while the dark AUC deficit is present only at the strictest cutoff. Adjusted models show that darker skin is associated with more false positives and lighter skin with more false negatives, with lesion contrast moving both error types in opposite directions.

5. A quantitative XAI analysis showing that all three architectures place less attribution inside the lesion on darker images, and that false negatives have lower attribution mass than false positives, which explains the weak pooled mass-error correlation. The results are reported in Section 4.6.

6. An external validation on HIBA that reproduces the CNN-transformer gap, shows a 16 to 18 point sensitivity drop, and confirms that HIBA contains no Fitzpatrick IV melanomas and no Fitzpatrick V-VI images.

The training and fairness analysis use one public dataset, HAM10000. The lightness strata are image-based measurements, not clinical Fitzpatrick labels, and the results do not establish clinical utility. External validation on HIBA is reported in Section 4.7.

2 Related work

2.1 Melanoma classification and public datasets

Deep learning for skin lesion classification moved from dermatologist-level comparisons on single datasets [12] to multi-source benchmarks. HAM10000 contributed 10,015 dermoscopic images across seven diagnostic classes [35], and the ISIC challenges expanded the scale and the number of sources [6]. BCN20000 added 18,946 dermoscopic images with eight diagnostic classes [7], and the ISIC 2020 collection added patient-level context and a clinical endpoint [29]. Cassidy et al. analysed the ISIC datasets and showed that duplicate lesions, class imbalance, and inconsistent metadata affect benchmark comparisons [4]. Human-reader studies and reporting guidelines now set a higher bar for evaluation [36, 9]. Clinical translation also requires prospective validation and reporting standards that go beyond a single benchmark [26].

Architecture comparisons usually cover a ResNet-style CNN [18], an efficient CNN [34], and a vision transformer [11]. The comparisons differ in preprocessing, split protocol, and threshold choice, which makes cross-paper rankings difficult to interpret. This paper fixes the split, the seeds, and the operating point so that the architecture and the training condition are the only variables.

2.2 Class imbalance

Focal loss down-weights easy examples and is the most common loss-level remedy for class imbalance [21]. Class-balanced loss reweights classes by their effective number of samples [8], and class-balanced sampling changes the effective class prior directly. Johnson and Khoshgoftaar survey the broader set of resampling, reweighting, and augmentation methods [20]. Controlled studies in convolutional networks show that the best correction depends on the architecture and the dataset size [3]. Strong photometric and geometric augmentation is often added at the same time.

The three interventions are usually evaluated together, so their separate effects and their interactions with architecture are rarely reported. This matters because a correction that is neutral on average may help one architecture and hurt another. The present benchmark treats the correction as a pre-declared condition and tests the interaction directly.

2.3 Fairness in dermatology

Fitzpatrick17k provided the first large clinical image set with Fitzpatrick labels and showed performance gaps across skin types [15]. The Diverse Dermatology Images dataset added biopsy-proven clinical images with balanced skin tone representation [10]. SCIN added crowdsourced clinical images with self-reported and estimated Fitzpatrick labels [38]. Pipsqueak collected dermoscopic melanocytic lesions in darker skin and reported a large sensitivity drop on Fitzpatrick V-VI [30]. A recent review identifies fairness, generalizability, and replicability as the main unresolved problems in melanoma AI [24].

The dermoscopic datasets with Fitzpatrick labels remain small. HIBA contains 1,616 images from Argentina with Fitzpatrick labels [28], and PROVe-AI contains 603 biopsy-confirmed lesions [23]. Neither includes Fitzpatrick V-VI. PASSION provides clinical images from Sub-Saharan Africa with Fitzpatrick III-VI [14], but the imaging modality differs from dermoscopy. The available data therefore support an image-based lightness analysis on dermoscopy and a separate cross-modality analysis on clinical images, not a single clean Fitzpatrick comparison.

Group fairness also depends on which error is counted. Equal opportunity compares true positive rates across groups [17], while a false positive rate comparison captures the harm of unnecessary

biopsies. A model can improve one and worsen the other. This paper reports both and models them separately.

Shortcut learning complicates the interpretation of group gaps. Surgical skin markings can change melanoma predictions [39], and Bissoto et al. showed that several visual artifacts are exploitable and that current debiasing methods do not remove them [2]. Lesion-skin contrast is one such visual factor: it correlates with skin tone but also with lesion pigmentation. A fairness analysis that does not adjust for it can attribute a contrast effect to skin tone. This paper measures contrast from the lesion masks and includes it in the adjusted models.

2.4 Calibration

Thresholded metrics such as sensitivity at a fixed specificity depend on the score distribution, not only on the ranking quality. Modern neural networks are often miscalibrated [16], and calibration is central to clinical prediction models [37]. A correction that changes calibration can therefore change the sensitivity ranking while leaving the AUC ranking stable. This is the mechanism we test, rather than assume.

2.5 Explainable AI in dermatology

Class activation maps and Grad-CAM are the most used attribution methods in dermatology [40, 32]. Integrated Gradients and other axiomatic methods provide an alternative [33], and model-agnostic methods such as LIME and SHAP are used when gradients are unavailable [27, 22]. Clinician-facing studies show that explanations can improve trust and diagnostic accuracy [5], but the explanations are usually evaluated qualitatively.

Qualitative evaluation is not enough for a fairness claim. A heatmap that looks plausible can be insensitive to the model parameters [1], and explainable AI in health care has been criticized for promising more than it delivers [13]. This paper treats attribution as a measurement with a stated error type, not as evidence of clinical reasoning.

2.6 Quantitative XAI metrics

Attribution localization can be measured against lesion segmentation masks: the fraction of attribution mass inside the lesion, the pointing game, and the attribution IoU. Faithfulness can be measured with deletion and insertion curves [25], sensitivity-n [31], and the ROAD benchmark [19]. These metrics make it possible to test whether an explanation is clinically meaningful rather than visually plausible.

Dermatology studies rarely combine these metrics with skin tone stratification. The XAI analysis in this paper uses localization metrics across tone groups and error types, and it reports why faithfulness curves were not informative for these highly confident models.

2.7 Gap

Three questions remain open: whether imbalance correction changes the ranking of architecture families, whether calibration explains the change, and whether attribution metrics can detect a group-level problem before it appears in aggregate sensitivity. This paper addresses the first two with paired tests and calibration, and it addresses the third with tone-stratified localization metrics and a false-positive/false-negative split.

3 Methods

3.1 Data and splits

We use HAM10000, which contains 10,015 dermoscopic images from 7,470 lesions across seven diagnostic classes [35]. The task is binary: melanoma versus all other diagnoses. The dataset contains 1,113 melanoma images, 11.1% of the total.

We use a lesion-aware split so that images of the same lesion never appear in more than one split. The split is 70% training, 15% validation, and 15% test. The test set contains 1,527 images and 186 melanomas. The split is generated once and reused by every run.

3.2 Preprocessing and augmentation

Images are resized to 160 pixels for the Stage 1 screen and 224 pixels for the Stage 2 comparison, and normalized with ImageNet statistics. The standard augmentation pipeline applies horizontal and vertical flips, random 90-degree rotations, and mild color jitter. The strong pipeline increases the color jitter and adds affine transformations with scale, translation, and rotation. The strong regime is designed to simulate domain shift.

3.3 Architectures

We compare ResNet-50, EfficientNet-B0, and ViT-S/16. All models use ImageNet pretrained weights and a two-class output. The ViT uses dynamic image sizing so that it accepts the 224-pixel input. The three models cover a standard CNN, an efficient CNN, and a transformer.

3.4 Training

All models use AdamW with a learning rate of 3×10^{-4}, weight decay 1×10^{-4}, and a cosine schedule. The batch size is 64, the maximum number of epochs is 15, and early stopping uses a patience of 5 on validation balanced accuracy. Gradients are clipped to a norm of 1.0. Training uses mixed precision when a GPU is available.

3.5 Conditions

Stage 1 is a screen on EfficientNet-B0 at 160 pixels. It contains eight configurations: CE with a random sampler, CE with class-balanced sampling, weighted CE with a random sampler, focal loss with $\gamma = 2$ and a random sampler, focal loss with $\gamma = 2$ and class-balanced sampling, focal loss with $\gamma = 3$ and class-balanced sampling, and two strong-augmentation variants. Each configuration uses three seeds, for 24 runs.

Stage 2 compares three pre-declared conditions across the three architectures at 224 pixels:

1. CE baseline: cross-entropy loss, random sampler, standard augmentation.
2. Planned imbalance correction: focal loss with $\gamma = 2$, class-balanced sampling, standard augmentation.
3. Stage 1 winner: cross-entropy loss, random sampler, strong augmentation.

Each condition uses three seeds, for 27 runs. The Stage 2 conditions are fixed before the test results are seen.

3.6 Metrics and threshold

We report sensitivity, specificity, balanced accuracy, AUC, and sensitivity at 95% specificity. The operating threshold is selected on the validation split at 95% specificity, frozen, and applied once to the test split. Each run writes per-sample test predictions so that the same scores can be reused for fairness and XAI analysis.

3.7 Statistical analysis

We report the mean and standard deviation across three seeds. We add 1,000-sample bootstrap confidence intervals on test for sensitivity, specificity, and AUC. The bootstrap resamples images within a run and skips degenerate samples. The intervals describe test-set uncertainty. The seed standard deviation describes run-to-run variation.

3.8 Fairness analysis

HAM10000 has no Fitzpatrick labels. We therefore compute an image-based lightness proxy from non-lesional skin. The image is converted to CIELAB, the lesion mask is dilated by 11 pixels and excluded, and very dark pixels ($L^* < 10$) and very bright pixels ($L^* > 95$) are removed. The median L^* of the remaining skin pixels is the proxy.

We group images into light ($L^* > 70$), medium ($55 < L^* \leq 70$), and dark ($L^* \leq 55$). These thresholds are specific to the HAM10000 lightness distribution. They are image-based strata, not clinical Fitzpatrick labels.

We also computed the Individual Typology Angle (ITA), $\text{ITA} = \arctan((L^* - 50)/b^*) \times 180/\pi$. The ITA was unstable on 37% of images because the b^* channel was near zero or negative after dermatoscope color casts. The L^* proxy is therefore the primary analysis, and the ITA is not used for group comparisons.

For each Stage 2 run, we compute per-group sensitivity, specificity, false positive rate, false negative rate, and AUC. The group intervals use the same bootstrap procedure as the overall metrics.

Lesion contrast is computed from the same masks. The lesion color is the median CIELAB value inside the mask, and the surrounding skin is the median inside an 11-pixel ring after excluding $L^* < 10$ and $L^* > 95$. We define ΔL^* as skin minus lesion, so a positive value means the lesion is darker than the surrounding skin, and ΔE as the CIELAB 76 difference between the two median colors.

The main analysis uses the pre-declared cutoff $L^* \leq 55$. We repeat the group analysis at $L^* \leq 58$ and $L^* \leq 60$ to test whether the results depend on the cutoff. We also fit adjusted logistic models of false positive and false negative on perilesional L^*, ΔL^*, log lesion area, age, sex, and anatomical site. Continuous predictors are standardized; one standard deviation is 7.2 L^* units and 10.3 contrast units in the test set. Standard errors come from a cluster bootstrap over lesions with 300 replicates, so repeated images and seeds of the same lesion are not treated as independent. The p-values are descriptive and are not corrected for multiplicity across cutoffs and architectures.

We compute expected calibration error with 10 equal-width bins and the Brier score on the test split. Paired sensitivity comparisons resample test images with replacement and keep the condition pairing. Difference-in-differences compares the CE-to-focal change of one architecture with that of another. Paired intervals use 1,000 bootstrap replicates.

External validation uses HIBA [28], a dermoscopic dataset from Argentina with Fitzpatrick labels. We apply the frozen HAM10000 validation threshold and do not refit on HIBA. As a secondary comparison, we also report sensitivity at 95% specificity with a threshold fitted on

HIBA. This oracle threshold is an upper bound, not a deployable result. PROVe-AI [23] is a possible extension but was not evaluated here. A clinical dark-skin dataset, such as DDI [10] or PASSION [14], is left for future work, with the imaging modality difference stated as a limitation.

3.9 XAI analysis

For the CE-baseline run of each architecture, we compute Grad-CAM heatmaps on a tone-stratified test sample: all 43 dark images plus 150 randomly drawn light and 150 medium images per seed. We measure:

- Attribution mass inside the lesion: $M = \sum_{i \in \text{lesion}} h_i / \sum_i h_i$, where h_i is the normalized heatmap value at pixel i.
- Pointing game accuracy: the fraction of images where the maximum heatmap value falls inside the lesion mask.
- Attribution IoU: the intersection over union between the thresholded heatmap and the lesion mask.

We then compute the Spearman correlation between attribution mass and error status, and between attribution mass and lightness group. The correlations pool the three seeds, so the p-values treat repeated images as independent and are descriptive rather than confirmatory. The XAI analysis uses the same test split and the same per-sample predictions as the classification and fairness analyses.

We also implemented deletion and insertion curves, but the models are highly confident: the target-class probability stays near 1.0 for both the original and the perturbed images, so the curves do not separate heatmaps on this benchmark. We therefore report localization metrics and treat faithfulness curves as future work.

3.10 Reproducibility

Every run writes its configuration, git commit, seed, device, package versions, metrics, and per-sample predictions. The split is cached and reused. Stage 2 training used Kaggle GPUs. The calibration, XAI, and external evaluations used a local Apple Silicon GPU. The code, configs, and table generation scripts are available in the project repository.

4 Results

4.1 Stage 1: loss, sampling, and augmentation screen

Table 1 reports the Stage 1 screen on EfficientNet-B0 at 160 pixels. The validation winner is CE with strong augmentation, with a validation sensitivity of 0.556 ± 0.027 at 95% specificity. The CE baseline reaches 0.536 ± 0.016. The planned focal loss with class-balanced sampling reaches 0.502 ± 0.039, and focal loss with $\gamma = 3$ and class-balanced sampling reaches 0.449 ± 0.051.

The planned imbalance correction did not beat the CE baseline on validation. The difference between the CE baseline and the strong-augmentation variant is small relative to the seed standard deviation. The test columns in Table 1 are descriptive; the model selection used validation only.

Table 1: Stage 1 screen on EfficientNet-B0 at 160 pixels. Validation sensitivity is used for model selection. Test sensitivity is reported at the validation-selected 95% specificity threshold. Values are mean and standard deviation across three seeds.

Configuration	Val sens	Test sens (95% CI)	Specificity	Balanced acc	AUC
stage1_ce_random_strong	0.556 ± 0.027	0.513 (0.438 to 0.582)	0.943	0.728	0.901
stage1_ce_random	0.536 ± 0.016	0.530 (0.456 to 0.600)	0.948	0.739	0.897
stage1_ce_cb	0.521 ± 0.086	0.484 (0.411 to 0.554)	0.938	0.711	0.890
stage1_focal2_random	0.509 ± 0.041	0.487 (0.412 to 0.560)	0.946	0.717	0.894
stage1_focal2_cb	0.502 ± 0.039	0.480 (0.408 to 0.551)	0.943	0.712	0.897
stage1_wce_random	0.502 ± 0.043	0.493 (0.421 to 0.564)	0.937	0.715	0.896
stage1_focal2_cb_strong	0.483 ± 0.013	0.491 (0.421 to 0.558)	0.943	0.717	0.896
stage1_focal3_cb	0.449 ± 0.051	0.478 (0.405 to 0.553)	0.946	0.712	0.899

4.2 Stage 2: architecture comparison

Table 2 reports the Stage 2 comparison at 224 pixels. The three conditions are the CE baseline, focal loss with class-balanced sampling, and CE with strong augmentation.

At the CE baseline, ResNet-50 reaches a test sensitivity of 0.570 (95% CI 0.497 to 0.639), EfficientNet-B0 reaches 0.556 (0.480 to 0.624), and ViT-S/16 reaches 0.400 (0.329 to 0.471). The two CNNs lead the transformer by a wide margin.

Under focal loss with class-balanced sampling, ResNet-50 falls to 0.453 (0.379 to 0.523), EfficientNet-B0 falls to 0.477 (0.405 to 0.548), and ViT-S/16 rises to 0.482 (0.408 to 0.556). The correction helps the transformer by 8.2 points and hurts the CNNs by 11.7 and 7.9 points.

The architecture ranking therefore reverses in the point estimates. At the CE baseline the order is ResNet-50, EfficientNet-B0, ViT-S/16. Under focal loss with class-balanced sampling the order is ViT-S/16, EfficientNet-B0, ResNet-50.

Strong augmentation is close to neutral. ResNet-50 reaches 0.568 (0.498 to 0.637), EfficientNet-B0 reaches 0.563 (0.489 to 0.636), and ViT-S/16 reaches 0.435 (0.366 to 0.503). The intervals overlap with the CE baseline for the two CNNs. The paired CE-to-strong changes are +0.007 for EfficientNet-B0, −0.002 for ResNet-50, and +0.036 for ViT-S/16, and every interval includes zero.

Table 3 reports paired tests that resample test images and keep the condition pairing. The focal effect within a single architecture is not significant: −0.079 (95% CI −0.210 to 0.055) for EfficientNet-B0, −0.116 (−0.244 to 0.031) for ResNet-50, and +0.082 (−0.038 to 0.237) for ViT-S/16. The difference-in-differences is significant for each CNN against the ViT: −0.161 (−0.289 to −0.049, $p = 0.008$) for EfficientNet-B0 minus ViT-S/16 and −0.199 (−0.437 to −0.011, $p = 0.032$) for ResNet-50 minus ViT-S/16. The correction therefore has an architecture-dependent effect. The individual condition changes are too uncertain for a precise ranking claim.

4.3 AUC is stable while sensitivity ranking flips

The AUC ranking does not follow the sensitivity ranking. At the CE baseline, the AUC is 0.914 for EfficientNet-B0, 0.897 for ResNet-50, and 0.854 for ViT-S/16. Under focal loss with class-balanced sampling, the AUC is 0.891 for EfficientNet-B0, 0.884 for ResNet-50, and 0.872 for ViT-S/16. EfficientNet-B0 remains first and ViT-S/16 remains last in both conditions.

The stability of the AUC ranking suggests that the sensitivity reversal is specific to the thresholded operating point. Calibration and threshold placement are plausible mechanisms. Both metrics matter: a clinical screening claim depends on the thresholded sensitivity, and the AUC describes the ranking quality of the scores.

Table 2: Stage 2 comparison at 224 pixels. Test sensitivity is reported at the validation-selected 95% specificity threshold. Values are mean and standard deviation across three seeds.

Configuration	Val sens	Test sens (95% CI)	Specificity	Balanced acc	AUC
stage2_efficientnet_b0_ce_random_strong	0.521 ± 0.021	0.563 (0.489 to 0.636)	0.952	0.757	0.912
stage2_efficientnet_b0_ce_random	0.553 ± 0.035	0.556 (0.480 to 0.624)	0.953	0.754	0.914
stage2_efficientnet_b0_focal2_cb	0.481 ± 0.042	0.477 (0.405 to 0.548)	0.950	0.713	0.891
stage2_resnet50_ce_random	0.526 ± 0.044	0.570 (0.497 to 0.639)	0.951	0.761	0.897
stage2_resnet50_ce_random_strong	0.549 ± 0.015	0.568 (0.498 to 0.637)	0.942	0.755	0.899
stage2_resnet50_focal2_cb	0.489 ± 0.027	0.453 (0.379 to 0.523)	0.947	0.700	0.884
stage2_vit_s_focal2_cb	0.474 ± 0.034	0.482 (0.408 to 0.556)	0.940	0.711	0.872
stage2_vit_s_ce_random_strong	0.466 ± 0.030	0.435 (0.366 to 0.503)	0.941	0.688	0.866
stage2_vit_s_ce_random	0.423 ± 0.042	0.400 (0.329 to 0.471)	0.941	0.670	0.854

Table 3: Paired sensitivity comparisons. Differences are paired by test image; intervals are 95% bootstrap intervals over images. The within-architecture focal effects have intervals that include zero, the strong-augmentation changes are close to zero, and the difference-in-differences between each CNN and the ViT excludes zero.

Test	Architecture	Δ sensitivity	95% CI	p
Focal+CB vs CE	EfficientNet-B0	-0.079	-0.210 to +0.055	0.590
Focal+CB vs CE	ResNet-50	-0.116	-0.244 to +0.031	0.248
Focal+CB vs CE	ViT-S/16	+0.082	-0.038 to +0.237	0.440
CE+strong vs CE	EfficientNet-B0	+0.007	-0.173 to +0.177	0.526
CE+strong vs CE	ResNet-50	-0.002	-0.095 to +0.091	0.966
CE+strong vs CE	ViT-S/16	+0.036	-0.067 to +0.157	1.000
Difference in differences	EfficientNet-B0 minus ViT-S/16	-0.161	-0.289 to -0.049	0.008
Difference in differences	ResNet-50 minus EfficientNet-B0	-0.038	-0.273 to +0.137	0.936
Difference in differences	ResNet-50 minus ViT-S/16	-0.199	-0.437 to -0.011	0.032

4.4 Calibration tracks the sensitivity reversal

Table 4 reports test-set calibration at the validation-selected threshold. Focal loss with class-balanced sampling roughly triples the expected calibration error: 0.203 for ResNet-50, 0.178 for EfficientNet-B0, and 0.199 for ViT-S/16, against 0.046, 0.060, and 0.146 at the CE baseline. Strong augmentation keeps calibration close to the CE baseline, with ECE between 0.046 and 0.058. The Brier score follows the same pattern.

The ViT is the worst-calibrated CE baseline, with an ECE of 0.146, and strong augmentation reduces it to 0.058. The focal condition makes all three architectures worse calibrated. Because the operating threshold is selected on validation at 95% specificity, a miscalibrated score distribution moves the test operating point and changes sensitivity. This is consistent with the stable AUC and the unstable sensitivity ranking.

Figure 1 shows reliability curves and ECE by lightness group. The dark stratum has higher ECE than the light and medium groups for all three CE baselines: 0.126 against 0.072 and 0.052 for EfficientNet-B0, 0.132 against 0.063 and 0.035 for ResNet-50, and 0.235 against 0.144 and 0.149 for ViT-S/16.

Table 4: Test-set calibration at the validation-selected threshold. ECE uses 10 equal-width bins. Values are mean and standard deviation across three seeds.

Configuration	ECE	Brier
ResNet-50 / CE	0.046 ± 0.013	0.074 ± 0.004
ResNet-50 / CE+strong	0.046 ± 0.004	0.076 ± 0.004
ResNet-50 / Focal+CB	0.203 ± 0.041	0.133 ± 0.016
EfficientNet-B0 / CE	0.060 ± 0.009	0.076 ± 0.001
EfficientNet-B0 / CE+strong	0.052 ± 0.015	0.076 ± 0.003
EfficientNet-B0 / Focal+CB	0.178 ± 0.049	0.134 ± 0.022
ViT-S/16 / CE	0.146 ± 0.109	0.126 ± 0.041
ViT-S/16 / CE+strong	0.058 ± 0.037	0.089 ± 0.012
ViT-S/16 / Focal+CB	0.199 ± 0.039	0.141 ± 0.011

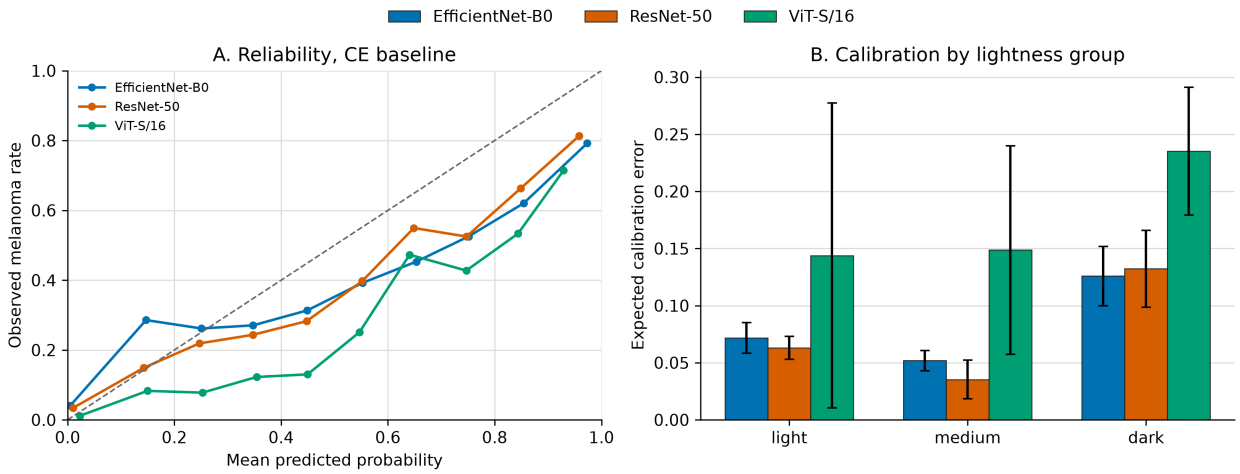


Figure 1: Calibration at the CE baseline. (A) Reliability curves for the three architectures. (B) Expected calibration error by lightness group, mean and seed standard deviation across the CE-baseline runs.

4.5 Fairness across lightness strata

The test set contains 644 light images with 81 melanomas, 840 medium images with 99 melanomas, and 43 dark images with 6 melanomas. The dark stratum is 2.8% of the test set and 3.2% of the melanoma cases.

Table 5 reports the per-group sensitivity at 95% specificity for the nine Stage 2 configurations at the pre-declared cutoff $L^* \leq 55$. Figure 2 shows the same cutoff per configuration. At this cutoff the dark stratum has the lowest AUC in all nine configurations, 0.71 to 0.87, against 0.83 to 0.90 for light and 0.88 to 0.93 for medium. It also has the highest false positive rate, 0.08 to 0.19, against 0.04 to 0.07 for the other two groups.

The cutoff is a choice, so we repeated the analysis at $L^* \leq 58$ and $L^* \leq 60$ (Table 6, Figure 3). The dark AUC deficit is not robust to the cutoff. At $L^* \leq 58$ the dark AUC is 0.891, against 0.868 for light and 0.908 for medium. At $L^* \leq 60$ it is 0.881, against 0.868 and 0.912. The dark false positive rate is elevated at every cutoff: 0.126 at $L^* \leq 55$, 0.086 at $L^* \leq 58$, and 0.073 at $L^* \leq 60$, against 0.047 to 0.050 for light and medium. Dark sensitivity is 0.500 with 6 melanomas, 0.648 with 12, and 0.580 with 23. The intervals are wide, and dark sensitivity is not consistently lower

Table 5: Lightness-stratified test sensitivity at 95% specificity. Values are mean and standard deviation across three seeds. The dark stratum contains 43 test images and six melanoma cases. These are image-based lightness groups, not clinical Fitzpatrick labels.

Configuration	Light sens	Medium sens	Dark sens	Dark AUC	Dark FPR	Dark n / pos.
EfficientNet-B0 / CE	0.477 ± 0.031	0.609 ± 0.050	0.722 ± 0.096	0.871	0.117	43 / 6
EfficientNet-B0 / CE+strong	0.519 ± 0.054	0.599 ± 0.086	0.556 ± 0.096	0.833	0.090	43 / 6
EfficientNet-B0 / Focal+CB	0.420 ± 0.064	0.522 ± 0.076	0.500 ± 0.167	0.806	0.081	43 / 6
ResNet-50 / CE	0.502 ± 0.036	0.633 ± 0.050	0.444 ± 0.096	0.803	0.117	43 / 6
ResNet-50 / CE+strong	0.527 ± 0.026	0.603 ± 0.038	0.556 ± 0.096	0.854	0.135	43 / 6
ResNet-50 / Focal+CB	0.399 ± 0.093	0.481 ± 0.066	0.722 ± 0.192	0.848	0.162	43 / 6
ViT-S/16 / CE	0.342 ± 0.091	0.451 ± 0.065	0.333 ± 0.167	0.740	0.144	43 / 6
ViT-S/16 / CE+strong	0.407 ± 0.054	0.471 ± 0.132	0.222 ± 0.096	0.763	0.126	43 / 6
ViT-S/16 / Focal+CB	0.362 ± 0.062	0.586 ± 0.010	0.389 ± 0.096	0.712	0.189	43 / 6

Table 6: Dark-skin threshold sensitivity at 95% specificity. Values are means across the nine CE-baseline runs. The dark AUC gap is present at $L^* \leq 55$ but not at $L^* \leq 58$ or 60, while the dark false positive rate remains elevated at every cutoff. These are image-based lightness groups, not clinical Fitzpatrick labels.

L^* cutoff	Group	n	Melanomas	Sensitivity	AUC	FPR
55	light	644	81	0.440	0.868	0.050
	medium	840	99	0.565	0.911	0.049
	dark	43	6	0.500	0.805	0.126
58	light	644	81	0.440	0.868	0.050
	medium	777	93	0.550	0.908	0.048
	dark	106	12	0.648	0.891	0.086
60	light	644	81	0.440	0.868	0.050
	medium	704	82	0.556	0.912	0.047
	dark	179	23	0.580	0.881	0.073

than light sensitivity.

The adjusted models in Table 7 separate false positives from false negatives. Darker skin is associated with more false positives: the L^* odds ratio is 0.78 for ResNet-50, 0.88 for EfficientNet-B0, and 0.77 for ViT-S/16 per standard deviation. The interval excludes 1 for ResNet-50; the other two include 1. Lighter skin is associated with more false negatives: the L^* odds ratio is 1.74, 1.66, and 1.75, and all three intervals exclude 1. Lesion contrast moves in the opposite direction for the two error types. Higher contrast increases false positives (OR 2.05, 1.69, and 2.42) and reduces false negatives (OR 0.58, 0.74, and 0.48). The pooled error model shows no L^* association, with odds ratios from 0.92 to 1.02 and all intervals including 1, because the two effects cancel.

This changes the fairness interpretation. The strict dark group has a higher false positive rate across cutoffs, but the AUC deficit is cutoff-dependent and dark sensitivity is not lower. The error profile shifts with tone: darker images produce more false positives, lighter images produce more false negatives. Lesion contrast is the stronger correlate of both.

4.6 Quantitative XAI

Table 8 reports attribution mass inside the lesion, pointing-game accuracy, attribution IoU, and error rate for the CE-baseline models. Table 9 reports the Spearman correlations. The sample contains 129 dark images and 18 melanomas across the three seeds, and 450 light and 450 medium

Table 7: Adjusted odds ratios for false positives and false negatives. Each model includes perilesional L^*, lesion contrast, log lesion area, age, sex, and anatomical site. Odds ratios are per one standard deviation; intervals are lesion-cluster bootstrap 95% CIs. The pooled error model is not shown because the L^* effects on false positives and false negatives have opposite signs.

Error type	Architecture	L^* OR (95% CI)	Contrast OR (95% CI)
False positive	ResNet-50	0.78 (0.61 to 0.94)	2.05 (1.66 to 2.68)
	EfficientNet-B0	0.88 (0.67 to 1.11)	1.69 (1.39 to 2.14)
	ViT-S/16	0.77 (0.57 to 1.03)	2.42 (1.94 to 3.14)
False negative	ResNet-50	1.74 (1.11 to 2.58)	0.58 (0.35 to 1.02)
	EfficientNet-B0	1.66 (1.16 to 2.34)	0.74 (0.51 to 1.13)
	ViT-S/16	1.75 (1.14 to 2.60)	0.48 (0.29 to 0.75)

Table 8: Quantitative XAI at the CE baseline on the HAM10000 test split. Each architecture uses three seeds and a tone-stratified sample of 343 test images per seed. Attribution mass is the fraction of the Grad-CAM heatmap inside the lesion mask. Values are means across seeds.

Architecture	Tone group	n	n positive	Attribution mass	Pointing game	Attribution IoU	Error rate
ResNet-50	light	450	36	0.593	0.882	0.404	0.076
	medium	450	54	0.501	0.809	0.362	0.078
	dark	129	18	0.504	0.674	0.330	0.178
EfficientNet-B0	light	450	36	0.456	0.718	0.333	0.076
	medium	450	54	0.374	0.613	0.284	0.091
	dark	129	18	0.342	0.449	0.208	0.140
ViT-S/16	light	450	36	0.341	0.604	0.277	0.109
	medium	450	54	0.251	0.367	0.173	0.116
	dark	129	18	0.265	0.248	0.154	0.217

images.

Attribution mass inside the lesion is highest in the light stratum for all three architectures. EfficientNet-B0 falls from 0.456 to 0.374 to 0.342 across light, medium, and dark. ViT-S/16 falls from 0.341 to 0.251 and then rises slightly to 0.265. ResNet-50 falls from 0.593 to 0.501 and then stays near 0.504. The mass-tone correlation is negative and significant for every architecture: -0.170 for EfficientNet-B0, -0.167 for ResNet-50, and -0.184 for ViT-S/16, all with $p < 10^{-7}$.

Pointing-game accuracy is lowest in the dark stratum in every case: 0.449 for EfficientNet-B0, 0.674 for ResNet-50, and 0.248 for ViT-S/16. Attribution IoU follows the same direction for EfficientNet-B0 and ViT-S/16, and is only slightly lower in the dark stratum than the medium stratum for ResNet-50 (0.330 vs 0.362).

Attribution mass separates the two error types. False negatives have lower mass inside the lesion than false positives for EfficientNet-B0 (0.285 vs 0.522) and ViT-S/16 (0.298 vs 0.402). ResNet-50 is similar for the two error types (0.578 vs 0.565). The pooled mass-error correlation is therefore near zero: false negatives move below the correct-class mass and false positives move above it. The XAI signal is group-level and error-type-specific, not a per-image predictor of error.

Figure 4 shows the group pattern. ResNet-50 localizes best in absolute terms and has the highest dark-stratum attribution mass, but its dark false positive rate is not the lowest. Localization quality and error rate move together across tone groups, but they do not rank architectures the same way.

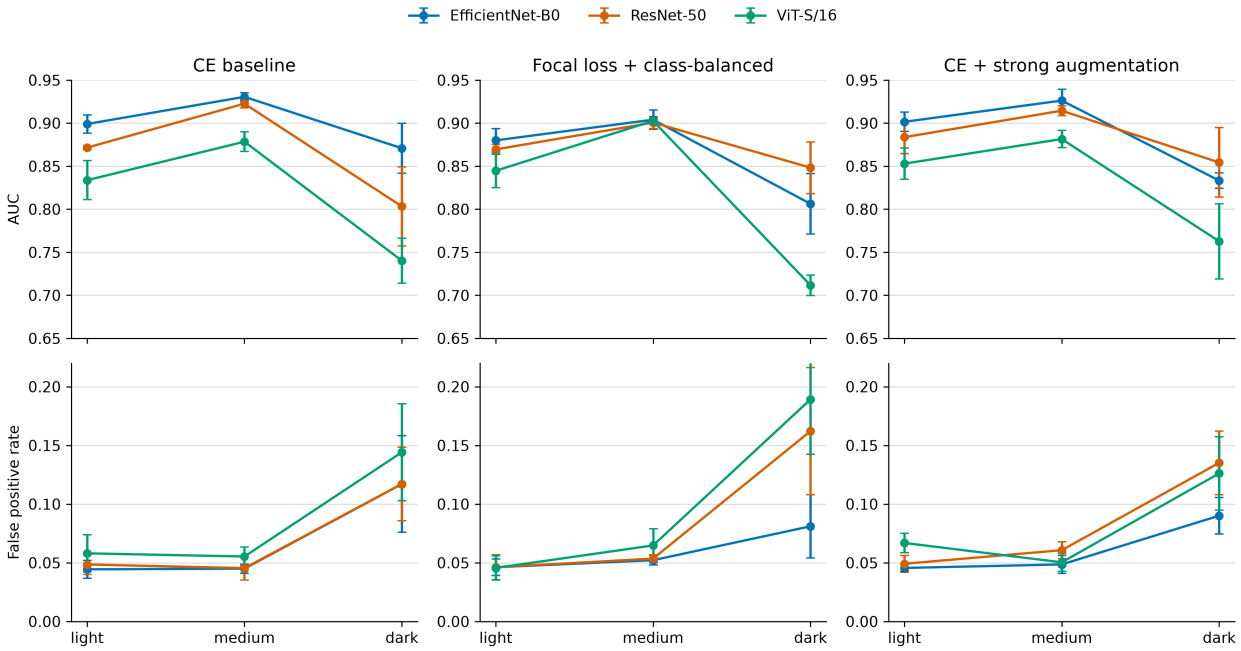


Figure 2: AUC and false positive rate by lightness group for the nine Stage 2 configurations at the pre-declared cutoff $L^* \leq 55$. Columns are training conditions and lines are architectures. Points are means across three seeds; error bars are seed standard deviations. The dark stratum contains 43 test images and six melanoma cases.

Table 9: Spearman correlations between attribution mass inside the lesion, lightness group, and error status at the CE baseline. Tone is coded light = 0, medium = 1, dark = 2. A negative mass-tone correlation means less attribution inside the lesion on darker images.

Architecture	Images	ρ (mass, tone)	p	ρ (mass, error)	p
EfficientNet-B0	1027	-0.170	4.6×10^{-8}	-0.023	0.458
ResNet-50	1029	-0.167	7.3×10^{-8}	0.047	0.131
ViT-S/16	1029	-0.184	3.0×10^{-9}	0.089	0.004

4.7 External tone-labeled validation

Table 10 reports the HIBA dermoscopic subset for the CE-baseline models: 1,280 images and 196 melanomas. We applied the HAM10000 validation threshold at 95% specificity without refitting. HIBA contains Fitzpatrick I-IV only, and 108 images have no Fitzpatrick label.

Sensitivity drops on the external set. ResNet-50 reaches 0.391 at 0.944 specificity, EfficientNet-B0 reaches 0.396 at 0.922, and ViT-S/16 reaches 0.221 at 0.967. The AUCs are 0.807, 0.782, and 0.791. On the HAM10000 test set the same models reached 0.57, 0.56, and 0.40 sensitivity at 95% specificity, with AUCs of 0.90, 0.91, and 0.85. Both CNNs outperform the transformer on external sensitivity, which matches the HAM10000 baseline, but the two CNNs are within half a point of each other. The absolute drop is consistent with domain shift between the Argentine HIBA collection and HAM10000.

The per-type estimates do not show a monotonic skin-tone gradient. For Fitzpatrick I, II, and III, CNN sensitivity ranges from 0.35 to 0.43, and the ViT is lower at 0.20 to 0.33. Fitzpatrick

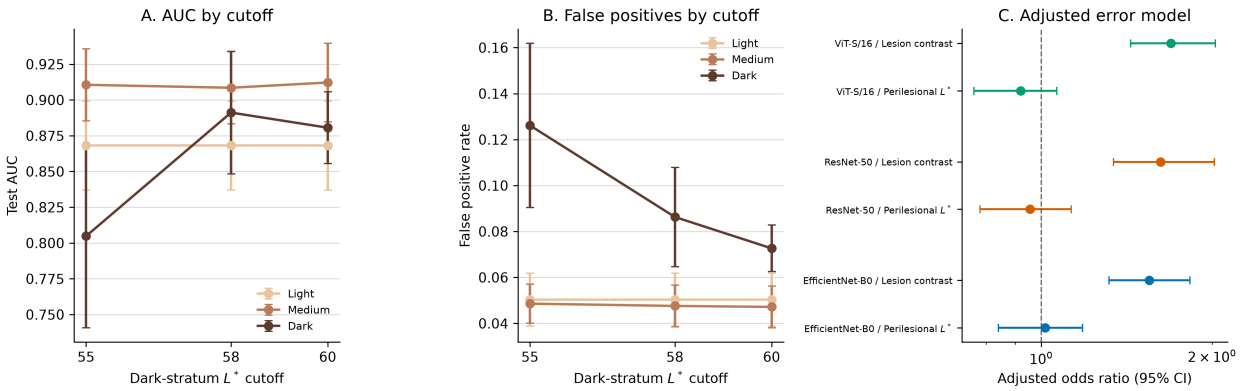


Figure 3: Dark-skin robustness. (A) AUC by L^* cutoff. (B) False positive rate by cutoff. (C) Adjusted odds ratios for the pooled error model, per standard deviation, with lesion-cluster bootstrap 95% CIs. The false-positive and false-negative models are reported separately in Table 7.

IV contains 25 images and no melanomas, so sensitivity is undefined and only specificity can be estimated. The group sizes are unbalanced: 804 of the 1,280 images are type II. HIBA cannot test the dark-skin claim, because it has almost no type IV melanoma cases and no type V-VI images at all.

Table 11 compares the CE baseline and the focal condition on HIBA. The focal condition reproduces the architecture interaction on sensitivity: ViT-S/16 rises from 0.221 to 0.315 (+9.4 points), while ResNet-50 falls from 0.391 to 0.315 (−7.6) and EfficientNet-B0 from 0.396 to 0.253 (−14.3). At the oracle 95% specificity threshold the direction is the same: ViT-S/16 rises from 0.310 to 0.335, while ResNet-50 falls from 0.366 to 0.245 and EfficientNet-B0 from 0.311 to 0.240. The interaction therefore transfers to a second dermoscopic source.

The external AUC ranking does not transfer. Under focal loss the external AUC ranking is ViT-S/16 0.794, ResNet-50 0.774, and EfficientNet-B0 0.722, while the in-domain AUC ranking at the same condition is EfficientNet-B0 0.891, ResNet-50 0.884, and ViT-S/16 0.872. The ViT gain is therefore a threshold effect rather than a ranking-quality gain: its AUC is essentially unchanged, from 0.791 to 0.794, and the sensitivity gain appears at the operating point. This is the same pattern as the in-domain calibration result.

The CNN-transformer gap transfers at the CE baseline and disappears under focal loss. The absolute performance does not transfer: the frozen HAM10000 threshold yields much lower sensitivity on HIBA, which means the operating point needs recalibration for a new source. The tone-specific claim remains limited to the image-based lightness strata in HAM10000 and to the error-type-specific associations reported above.

4.8 Figures

Figure 5 shows the training curves for the median seed of the best Stage 2 configuration by mean test sensitivity, ResNet-50 with the CE baseline. Figure 6 shows the corresponding test ROC curve.

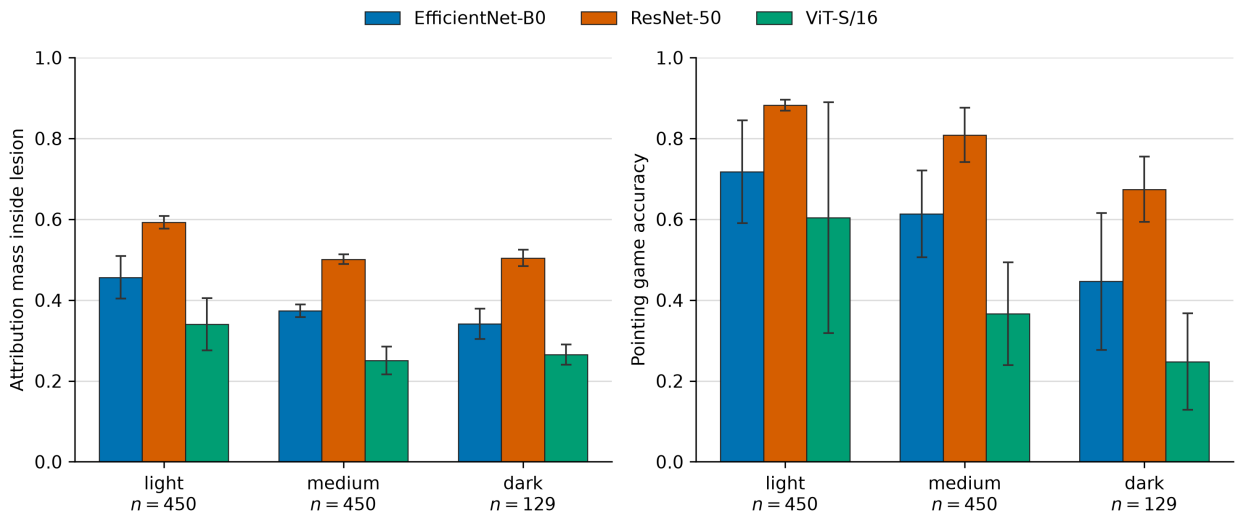


Figure 4: Attribution mass inside the lesion and pointing-game accuracy by lightness group for the three CE-baseline architectures. Bars are means across three seeds; error bars are seed standard deviations. The sample contains all 43 dark test images per seed and 150 randomly drawn light and medium images per seed.

5 Discussion

5.1 Imbalance correction is architecture-dependent

Focal loss with class-balanced sampling is not a neutral intervention. It improves the transformer and reduces both CNNs in the point estimates. A benchmark that applied the correction to one architecture and generalized the result would have reached the wrong conclusion about the others.

The paired tests sharpen this claim. The change within a single architecture is not significant: the 95% intervals for EfficientNet-B0, ResNet-50, and ViT-S/16 all include zero. The difference-in-differences is significant for each CNN against the ViT, at -0.161 for EfficientNet-B0 minus ViT-S/16 and -0.199 for ResNet-50 minus ViT-S/16. The robust finding is therefore an interaction between architecture and correction, not a precise ranking of the three models. The point estimates reverse, but three seeds do not support a narrow effect size.

The effect has a plausible mechanism. Focal loss changes the gradient distribution, and class-balanced sampling changes the effective class prior. A transformer with weaker inductive bias may benefit from the harder examples and the reweighted prior, while a CNN with strong local features may overfit the minority samples or lose calibration. The experiment cannot separate those mechanisms. We report the effect as an empirical interaction, not a causal explanation.

5.2 Calibration tracks the sensitivity reversal

The sensitivity ranking reverses, but the AUC ranking does not. EfficientNet-B0 remains first and ViT-S/16 remains last in both the baseline and corrected conditions. The reversal is therefore specific to the thresholded operating point.

The calibration results identify a mechanism for the thresholded instability, but not for the direction of the architecture effect. Focal loss with class-balanced sampling roughly triples the expected calibration error, from 0.046 to 0.203 for ResNet-50, from 0.060 to 0.178 for EfficientNet-

Table 10: External validation on the HIBA dermoscopic subset for the CE-baseline models (1,280 images, 196 melanomas). The threshold is the HAM10000 validation threshold at 95% specificity, frozen before the external evaluation. Values are means across three seeds. Fitzpatrick IV contains 25 images and no melanomas, so sensitivity is undefined. The 108 images without a Fitzpatrick label are included in the All row only.

Architecture	Fitzpatrick	n	Melanomas	Sensitivity	Specificity	AUC
ResNet-50	I	97	25	0.360	0.954	0.679
	II	804	144	0.403	0.922	0.780
	III	246	17	0.412	0.978	0.946
	IV	25	0	–	1.000	–
	All	1280	196	0.391	0.944	0.807
EfficientNet-B0	I	97	25	0.427	0.940	0.741
	II	804	144	0.389	0.890	0.745
	III	246	17	0.353	0.974	0.860
	IV	25	0	–	0.987	–
	All	1280	196	0.396	0.922	0.782
ViT-S/16	I	97	25	0.200	0.995	0.773
	II	804	144	0.227	0.950	0.777
	III	246	17	0.333	0.994	0.849
	IV	25	0	–	1.000	–
	All	1280	196	0.221	0.967	0.791

B0, and from 0.146 to 0.199 for ViT-S/16. Strong augmentation keeps ECE between 0.046 and 0.058. Since the operating threshold is selected on validation at 95% specificity, a miscalibrated score distribution moves the test operating point and changes sensitivity without changing the AUC ranking. Why the same correction helps the ViT and hurts the CNNs remains unexplained.

This matters for how model comparisons are reported. A single threshold at 0.5, or a single sensitivity number, can make an architecture look better or worse depending on the correction. Reporting sensitivity at a fixed specificity, AUC, and calibration together gives a more complete picture. The calibration analysis should be part of the model selection record, not an afterthought.

5.3 The fairness signal is error-type-specific

At the pre-declared cutoff $L^* \leq 55$, the dark stratum has the lowest AUC and the highest false positive rate in all nine configurations. The AUC result does not survive a change of cutoff: at $L^* \leq 58$ and $L^* \leq 60$ the dark AUC is close to the light AUC and below the medium AUC. The false positive elevation does survive. The dark false positive rate is 0.126 at $L^* \leq 55$, 0.086 at $L^* \leq 58$, and 0.073 at $L^* \leq 60$, against 0.047 to 0.050 for light and medium at every cutoff. The dark sensitivity estimate is not consistently lower than light sensitivity at any cutoff, and it is too uncertain to interpret.

The adjusted models explain why a single error metric hides the pattern. Darker skin is associated with more false positives, with L^* odds ratios of 0.78, 0.88, and 0.77 for the three architectures. Lighter skin is associated with more false negatives, with odds ratios of 1.74, 1.66, and 1.75. The false-negative association is significant for all three architectures; the false-positive association is

Table 11: External validation on HIBA for the CE-baseline and focal-loss models. Thresholds are the frozen HAM10000 validation thresholds. Sensitivity at 95% specificity uses an oracle threshold fitted on HIBA and is an upper bound, not a deployable result. Values are means across three seeds.

Architecture	Condition	Sensitivity	Specificity	AUC	Sens. at 95% spec.
ResNet-50	CE	0.391	0.944	0.807	0.366
ResNet-50	Focal+CB	0.315	0.925	0.774	0.245
EfficientNet-B0	CE	0.396	0.922	0.782	0.311
EfficientNet-B0	Focal+CB	0.253	0.945	0.722	0.240
ViT-S/16	CE	0.221	0.967	0.791	0.310
ViT-S/16	Focal+CB	0.315	0.954	0.794	0.335

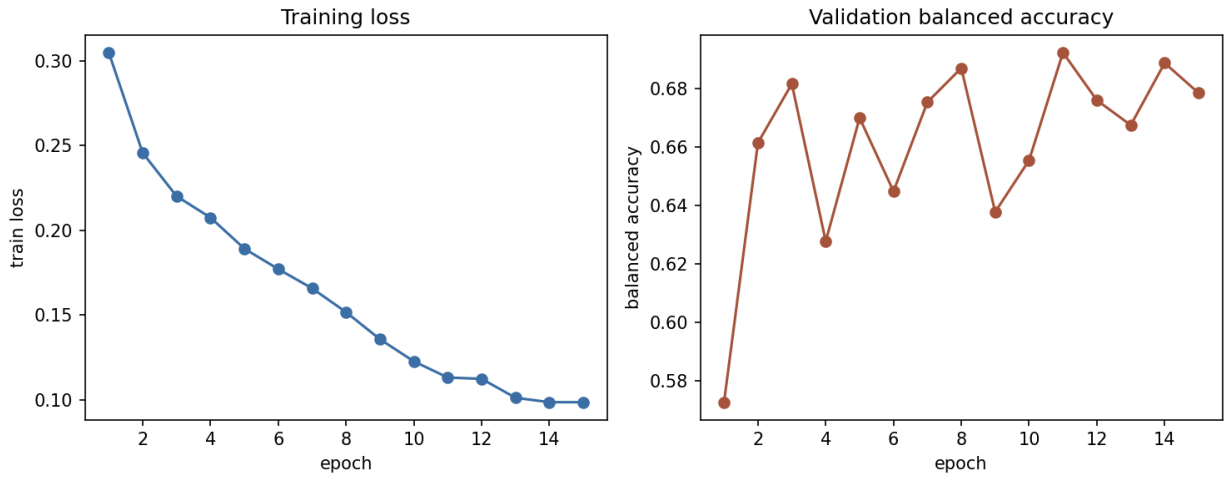


Figure 5: Training loss and validation balanced accuracy for the median seed of ResNet-50 with the CE baseline. The model trains for 15 epochs and restores the best validation checkpoint.

significant for ResNet-50 and directional for the other two. Lesion contrast moves in the opposite direction for the two error types: higher contrast increases false positives and reduces false negatives. The pooled error model shows no L^* association because the two effects cancel.

This is a more precise fairness claim than a global sensitivity gap. The model does not simply perform worse on darker skin. It shifts its operating behaviour: darker images generate more false alarms, while lighter images generate more missed melanomas. A clinical evaluation that reports only sensitivity or only AUC can miss the shift.

The lesion-contrast result also changes the interpretation of the light stratum. Pale lesions on pale skin have low contrast, and the model is more likely to miss them. Dark lesions on lighter skin have high contrast, and the model is more likely to call them malignant even when they are benign. Skin tone and lesion contrast are correlated, but the adjusted models show that the error-type shift is better described by the lesion-skin contrast than by absolute lightness alone.

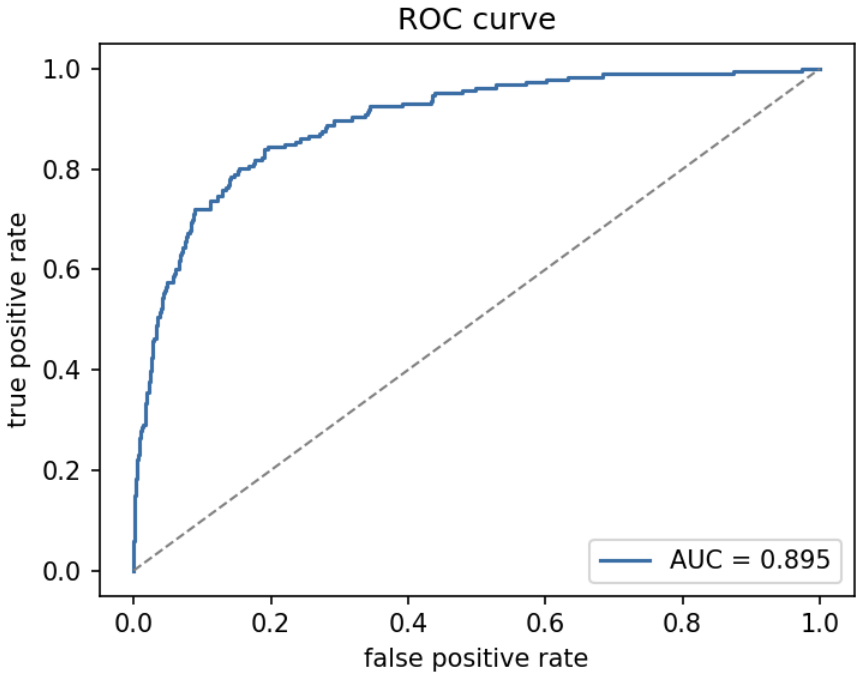


Figure 6: Test ROC curve for the median seed of ResNet-50 with the CE baseline. The operating threshold is selected on validation at 95% specificity.

5.4 The data bottleneck

The external dermoscopic datasets with Fitzpatrick labels, HIBA and PROVe-AI, contain Fitzpatrick I-IV only. The HIBA evaluation makes the constraint concrete. In 1,280 dermoscopic images, Fitzpatrick IV accounts for 25 images and not a single melanoma. Sensitivity is estimable only for types I-III, and those estimates show no monotonic skin-tone gradient. The 16 to 18 point sensitivity drop between HAM10000 and HIBA also shows that the available external set is a different imaging domain, not a clean replication. The external condition comparison does reproduce the architecture interaction: focal loss helps the ViT and hurts the CNNs on HIBA as well. That strengthens the benchmark claim, but it does not fill the Fitzpatrick IV-VI gap.

The clinical datasets with Fitzpatrick V-VI, such as DDI, SCIN, and PASSION, differ from HAM10000 in imaging modality. This is the main obstacle to a strong dark-skin claim. This study can quantify the representation gap, report the persistent false-positive elevation and the error-type shift, and show the limitation of the available dermoscopic data. A precise dark-skin sensitivity estimate requires new dermoscopic data collection or a carefully controlled cross-modality study.

5.5 What XAI adds

The XAI results show a group-level pattern that the aggregate metrics do not capture. All three architectures place less attribution inside the lesion on darker images, and the dark stratum has the highest error rate in every architecture. The direction is consistent across a standard CNN, an efficient CNN, and a transformer, so it is not a quirk of one inductive bias.

The error-type split gives the pattern a clearer interpretation. False negatives have lower attribution mass inside the lesion than false positives for EfficientNet-B0 (0.285 vs 0.522) and ViT-S/16

(0.298 vs 0.402). ResNet-50 is similar for the two error types (0.578 vs 0.565). False positives therefore look like attention to a benign lesion, and false negatives look like attention outside the melanoma. The pooled mass-error correlation is near zero because the two error types move in opposite directions.

This sets a limit on what attribution can claim. A deployment system should not use this heatmap score as a per-image triage signal without further work. It can use the group-level pattern as a monitoring signal: when a cohort shows lower attribution mass inside the lesion, the model may be missing more lesions in that cohort, while higher mass may indicate more false alarms. The direction of the change matters more than the absolute value.

ResNet-50 is the counterexample to a simple assumption that more attribution inside the lesion means fewer errors. It has the highest attribution mass and pointing-game accuracy in the dark stratum, but its dark false positive rate is not the lowest. Heatmaps are a partial view of model behaviour, not a substitute for group-level performance measurement.

5.6 Implications for model selection

The choice of architecture depends on the correction, but the paired intervals are wide. At the CE baseline, ResNet-50 has the highest point-estimate sensitivity at 95% specificity. Under focal loss with class-balanced sampling, ViT-S/16 has the highest point estimate. A deployment decision that ignores the training condition can pick a different model, but the current evidence supports an architecture-dependent effect rather than a precise ranking.

For the fairness analysis, the evaluation should include the architecture-condition combinations, the calibration, and the two error types. The best model on aggregate sensitivity may not have the lowest false positive rate in the dark stratum, and a model with a high AUC may still shift its errors in a way that matters clinically. The external validation adds a second source, but the available Fitzpatrick labels do not reach the darkest groups.

6 Limitations

- The training data are HAM10000 only. HIBA adds one external dermoscopic source, but cross-dataset generalization to PH2, BCN20000 [7], or ISIC 2020 is not measured.
- The external validation covers the CE baseline and focal loss. The strong-augmentation models are not evaluated on HIBA.
- The endpoint is binary melanoma versus all other diagnoses. Seven-class classification is a secondary experiment.
- Each condition uses three seeds. The paired within-architecture intervals include zero, so the paper supports an architecture-dependent interaction rather than a precise ranking.
- Stage 1 uses 160-pixel images and Stage 2 uses 224-pixel images. The two stages are not directly comparable.
- The fairness analysis uses an image-based lightness proxy, not clinical Fitzpatrick labels. The proxy is sensitive to illumination, color calibration, and lesion contrast.
- The pre-declared dark stratum contains 43 test images and six melanomas. The cutoff sensitivity analysis adds groups with 12 and 23 melanomas, but all dark-stratum estimates remain wide.

- The adjusted false-positive and false-negative models are observational. They identify associations, not causal effects, and the false-negative model has only 186 melanoma images across three seeds.
- HIBA and PROVe-AI contain Fitzpatrick I-IV only. HIBA has no Fitzpatrick IV melanomas, and the clinical datasets with Fitzpatrick V-VI differ in imaging modality, so the dark-skin comparison is confounded.
- Deletion and insertion curves saturated at a target probability near 1.0, so they could not separate heatmaps. Faithfulness is therefore not measured, and the XAI results cover localization only. A second attribution method is not included.
- No clinician reader study was conducted, so the clinical relevance of the attribution maps is not validated with dermatologists.
- No deployed system, prospective validation, or regulatory analysis is included.

7 Ethics and reproducibility

All data are public, de-identified research datasets. HAM10000 and the ISIC archive are released for research use, and the external datasets are used under their respective licenses. The study involves no patient contact and no new data collection.

The system is a research prototype, not a diagnostic device. The results do not establish clinical utility.

The Fitzpatrick scale has known limitations: it is a coarse phototype scale, it does not capture the full range of skin tones, and it conflates pigmentation with UV response. We use it because the datasets and the fairness literature use it. The lightness strata are image-based measurements, not clinical labels.

The representation gap is a finding with an ethical consequence: the data needed to measure dark-skin performance are scarce. We report the small number of dark-skin dermoscopic images. Synthetic dark-skin images are not used to claim fairness improvements.

The code, configs, splits, per-sample predictions, and table generation scripts are released with the paper. The repository records the git commit, seed, device, and package versions for every run.

Author contributions

N.Z. designed the study, implemented the code, ran the experiments, and wrote the manuscript. S.B. supervised the project and revised the manuscript. Both authors have read and agreed to the published version of the manuscript.

Institutional review board statement

Not applicable. All data are public, de-identified research datasets, and the study involved no patient contact or new data collection.

Informed consent statement

Not applicable.

Data availability

HAM10000 is available from the ISIC archive. HIBA is available from the ISIC archive under a CC-BY license. The per-sample test predictions, contrast features, fairness summaries, XAI metrics, and external-validation results are released with the code repository.

Code availability

The code is at `https://github.com/conorkz/melanoma-xai-bias`. The repository contains the training configs, split construction, evaluation scripts, table generators, and tests. The results in this paper were generated from the committed version of that repository.

Funding

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Conflicts of interest

The authors declare no conflicts of interest.

8 Conclusion

We benchmarked ResNet-50, EfficientNet-B0, and ViT-S/16 on HAM10000 under three pre-declared conditions. Focal loss with class-balanced sampling raised ViT-S/16 sensitivity by 8.2 points and lowered the two CNNs, which reversed the point-estimate ranking at 95% specificity. The paired within-architecture intervals include zero, but the difference-in-differences between each CNN and the ViT excludes zero. The AUC ranking stayed stable, and focal loss roughly tripled expected calibration error. Calibration and threshold placement therefore help explain how the sensitivity ranking can change while the ranking quality of the scores does not. The direction of the effect still depends on the architecture.

The fairness analysis became more specific once the cutoff and the error type were separated. The dark stratum has a higher false positive rate at L^* cutoffs of 55, 58, and 60, while the dark AUC deficit is present only at the strictest cutoff. Adjusted models show that darker skin is associated with more false positives and lighter skin with more false negatives, and that lesion-skin contrast moves both error types in opposite directions. Fairness in this setting is an error-type shift, not a single sensitivity gap.

The XAI analysis showed that all three architectures place less attribution inside the lesion on darker images. False negatives have lower attribution mass than false positives, which explains why the pooled mass-error correlation is weak. HIBA validation reproduced the interaction: focal loss raised ViT-S/16 sensitivity by 9.4 points and lowered the two CNNs by 7.6 and 14.3, while absolute CE-baseline sensitivity remained 16 to 18 points below HAM10000. HIBA contains no Fitzpatrick IV melanomas and no Fitzpatrick V-VI images.

The next steps are a clinical dark-skin evaluation with DDI or PASSION, two additional seeds for the six focal and CE cells, and a second attribution method. The clinical dataset is the main requirement for a precise Fitzpatrick V-VI claim. The additional seeds and the second attribution method would tighten the architecture interaction and test whether the error-type pattern is specific to Grad-CAM.

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